Why a polyhedron cannot have an odd number of identical faces

A guided proof

Fable 5.2 / prompting by Dan Horwitz

13 November 2026

> **About this guide.** It explains, step by step, a proof that was found in November 2026. You need vectors and the dot product, matrix multiplication, and complex numbers, as taught in a first university course. Everything else is built here, with a few exceptions, each marked "on trust" where it is used: Euler's formula for polyhedra, three facts about curves and pieces on the surface of a ball, and two school theorems about quadrilaterals and circles.
>
> The proof was found by an AI system (Claude Fable 5.2, made by Anthropic) prompted by the person named above. **No human mathematician has confirmed it yet.** If you find a step that does not follow, please report it through the repository that holds this file.

1. The question

A cube has six faces, and they are all the same square. A regular tetrahedron has four identical triangles. Figure 1 shows two more solids whose faces are all alike: one made of ten kites and one made of twelve rhombi.

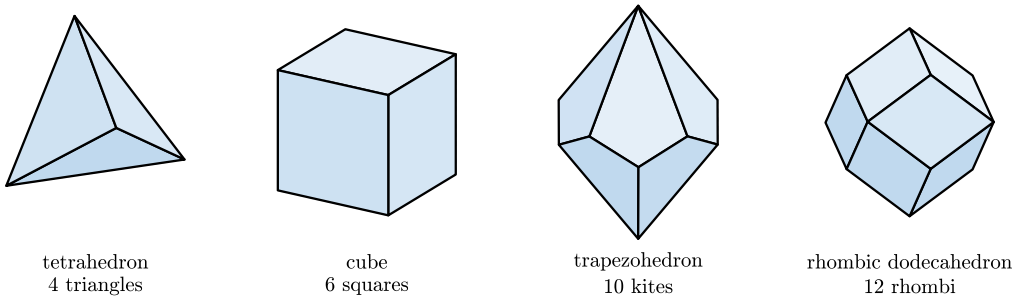


Figure 1: Four polyhedra with identical faces. Every one has an even number of them.

Four, six, ten, twelve. Try to find a polyhedron with five identical faces, or seven, or nine. Nobody ever has. For convex polyhedra, the ones without dents, the result that it cannot be done is attributed to a 1960 paper of Branko Grünbaum. For polyhedra with dents the question stayed open. In 2025 Nina Hungerbühler, Norbert Hungerbühler and Marcel Pirron showed that seven identical faces are impossible, and wrote that the general question was open. This guide explains why the answer is no.

Theorem. Take a closed surface without holes that is built from flat polygons, glued edge to edge, with at least three of them around every corner point. If the polygons are all congruent, then their number is even.

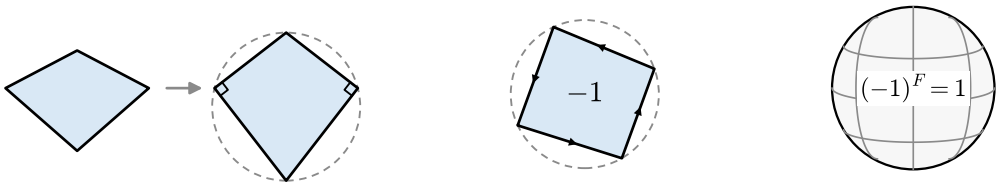
Congruent means: the same shape and the same size, where a mirror image counts as the same. The theorem allows a great deal.

- The surface may have dents. It may even pass through itself.
- It may live in space of any dimension, not only three.
- A polygon may have a corner with an angle of 180°, so that it looks as if it had one corner fewer. Such a corner still counts as a corner, and every face must then have it in the same place.

One condition cannot be dropped: *without holes*. In four-dimensional space there is a surface shaped like the skin of a doughnut that is built from nine identical squares. So any proof has to use, somewhere, that the surface is like a ball and not like a doughnut. Seeing where is one of the pleasures of this proof.

1.1. The plan

Counting edges settles most shapes of face in a few lines (Section 3). What survives are four-sided faces of two kinds: parallelograms and kites. Parallelograms have a short proof with strips of faces (Section 4). Kites are the hard case, and they take up the rest of the guide. The idea for them has three steps (Figure 2).



1. stretch until the corners of every face lie on a circle
2. going round such a face gives the sign −1
3. on a ball the signs multiply to +1

Figure 2: The plan for kites.

1. Stretch the polyhedron in a new direction until the four corners of every face lie on a circle (Section 9).
2. A face whose corners lie on a circle has a *sign*, and the sign is −1 (Section 6 and Section 10).
3. On a surface without holes the signs of all the faces multiply to +1 (Section 8).

So $(-1)^F = +1$, where F is the number of faces, and F is even. One warning now: for some shapes of kite the "stretch" of step 1 would have to be a shrinking, and it can then be carried out only in the formulas, not in a picture. The formulas are the same in both cases (Section 9.5).

2. Polyhedra as nets on a ball

First, what exactly is a polyhedron here? Think of a rubber ball with a *net* drawn on it: finitely many points, the *vertices*, joined by curves, the *edges*, that do not cross each other. The edges cut the surface of the ball into pieces, the *faces*. We ask that the net is connected and that every face is a disc, a piece without holes. We write V, E and F for the numbers of vertices, edges and faces.

A *polyhedron* is such a net together with a position in space for every vertex, chosen so that the corners of each face are the corners of a flat polygon whose boundary does not cross itself. We write X_v for the position of the vertex v. The cube is the net with $V = 8$, $E = 12$, $F = 6$ and the eight corner positions you expect. The ball is only a bookkeeping device. It records which faces are neighbours. The actual surface in space may be dented or may cut through itself, but its net is still drawn on a ball, and that is what "without holes" means.

Three facts about nets on a ball will be used again and again. The first is clear. The other two we take on trust.

- **Two faces at every edge.** Every edge has one face on each of its sides.
- **Opposite directions.** Walk round every face counterclockwise, as seen from outside the ball. Then every edge is walked twice, once by the face on each side, and the two walks go in opposite directions (Figure 3).
- **Euler's formula.** $V - E + F = 2$.

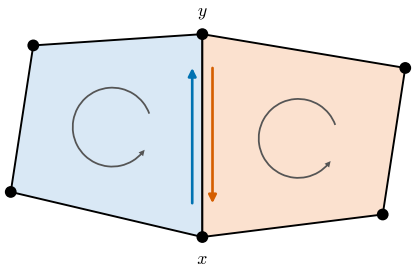


Figure 3: Two neighbouring faces, each walked counterclockwise. The shared edge is walked once from x to y and once from y to x.

3. Counting

Suppose a polyhedron has F congruent faces, each with n sides. Count the sides of all the faces: there are nF of them. Every edge is a side of exactly two faces, so

$$nF = 2E.$$

If n is odd, then F must be even. So polyhedra built from identical triangles, or from identical pentagons, always have an even number of faces. That is the whole proof for those shapes.

The same count works for one side length at a time. If the polygon has m sides of length ℓ, then mF is twice the number of edges of length ℓ. So if some length occurs an odd number of times round the polygon, F is even again.

From now on suppose that F is odd. We must reach a contradiction. By the count, every side length occurs an even number of times round the face. In particular n is even.

3.1. Only four sides

At least three faces meet at every vertex, so at least three edges end there. Every edge has two ends, so $2E \geq 3V$. Put this and $nF = 2E$ into Euler's formula:

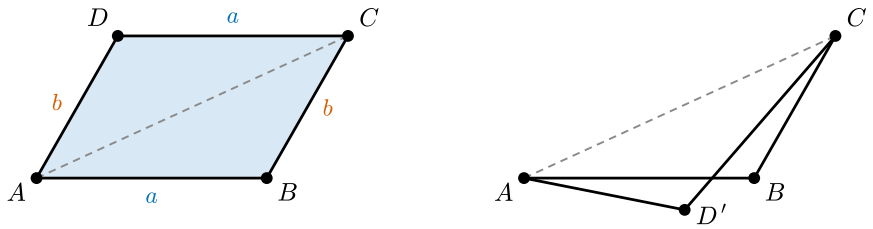
$$2 = V - E + F \leq \frac{2E}{3} - E + F = F - \frac{E}{3} = F - \frac{nF}{6} = F\frac{6-n}{6}.$$

The left side is positive, so $n < 6$. Since n is even and a polygon has at least three sides, $n = 4$. **Every face is a quadrilateral.**

3.2. Only two patterns

The four side lengths come in pairs. Going round the quadrilateral they read either a, b, a, b, where equal sides are opposite, or a, a, b, b, where equal sides are neighbours. If all four sides are equal, both descriptions fit, and we use the first.

The pattern a, b, a, b gives a parallelogram. Call the corners A, B, C, D. The diagonal AC cuts the quadrilateral into two triangles with the same three side lengths. Given A, B and C, that leaves two places for D (Figure 4). One makes a parallelogram. The other, D', is the mirror image of B in the line that cuts AC in half at a right angle. Then AB and CD' are mirror images of each other, and they cross on the mirror line, or else BC and $D'A$ do. If $a = b$, then B lies on the mirror line itself, D' is B, and the figure collapses. Neither is allowed for a face. So the face is a parallelogram.



a parallelogram — the same four sides, crossed

Figure 4: Sides a, b, a, b on a given diagonal: a parallelogram, or a crossed quadrilateral.

The pattern a, a, b, b with $a \neq b$ gives a kite. Name the corners as in Figure 5: A where the two sides of length a meet, T where the two sides of length b meet, and S and U for the other two. Both A and T are equally far from S and from U, so they lie on the line of all such points, the perpendicular bisector of SU. That line passes through the midpoint M of SU. So M lies on the line through A and T, which is called the *axis* of the kite, and we can write

$$M = A + \frac{r}{2}(T - A)$$

for some number r. This number describes the shape of the face. For $0 < r < 2$ the point M lies between A and T, and the face is an ordinary kite. For $r < 0$ or $r > 2$ it lies outside, and the face is a *dart*, a kite with a dent. For $r = 0$ or $r = 2$ the point M is A or T itself: the face is really a triangle, with an extra corner in the middle of its base.

Since $M = \frac{1}{2}(S + U)$, the formula for M can be rewritten without M:

$$S + U = (2 - r)A + rT. \tag{1}$$

We call (1) the *kite rule*. It holds for the corner positions of every face, with the same r, because the faces are congruent.

Three cases are left: parallelograms (Section 4), kites and darts with r different from 0 and 2 (Section 5 to Section 10), and the triangles with an extra corner (Section 11).

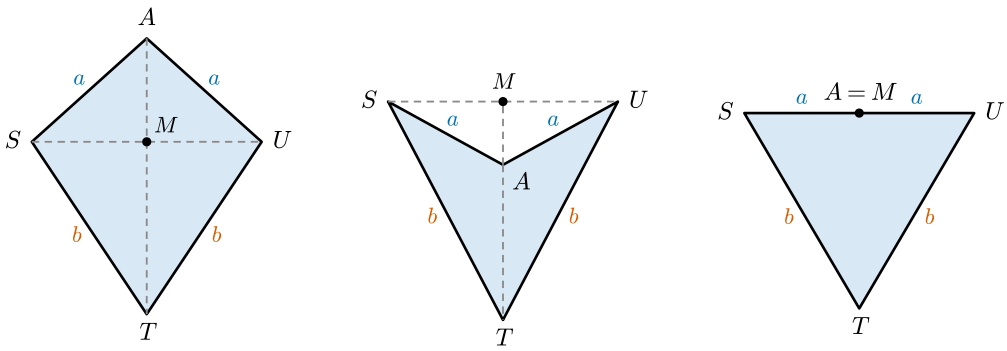
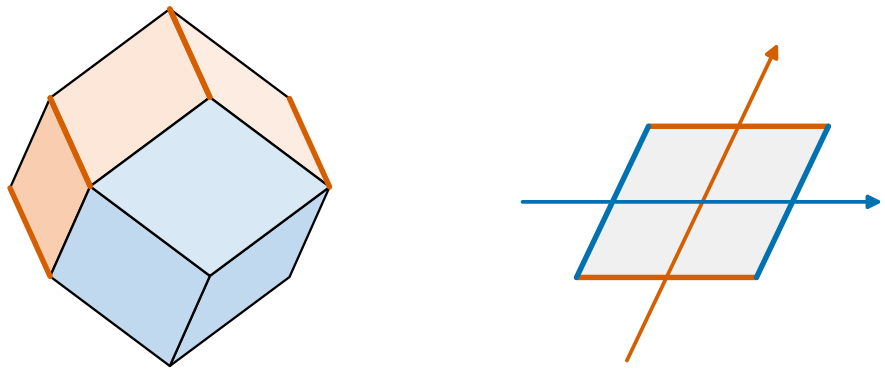


Figure 5: The faces with sides a, a, b, b, and their labels. M is the midpoint of SU.

4. Parallelograms

Let all faces be congruent parallelograms. In a parallelogram every side has an opposite side, parallel to it and of the same length. Every edge of the polyhedron lies in two faces, so it has two *partners*: its opposite side in the one face, and its opposite side in the other. Start at any edge, step to one of its partners, from there to that edge's other partner, and so on. There are finitely many edges, so at some moment the walk meets an edge for the second time. The walk can be traced backwards in only one way, so the first edge to be met again is the one where the walk started. The faces passed on the way form a closed strip, called a *zone* (Figure 6). All the edges that a zone crosses are parallel to each other. Every edge is crossed by exactly one zone, and only once.



a zone: the faces that share the thick parallel edges

through every face pass two zones, one for each pair of opposite sides

Figure 6: Zones on the rhombic dodecahedron.

Through every face pass two zones, one for each pair of opposite sides. **They are two different zones.** If one zone came through the same face by both pairs of sides, then all four sides of that face would be parallel to each other, and a parallelogram is not that flat.

Now draw each zone as a curve on the ball, running through the middle of its faces from one crossed edge to the next. The curve is a closed loop, and by what we just said it never crosses itself. Such a loop cuts the ball into two pieces, an inside and an outside; this is the second fact

about the ball that we take on trust. Take a second zone. It is a closed loop too. Each time it crosses the first loop it passes from the inside to the outside or back, and to return to its starting point it has to cross an even number of times (Figure 7).

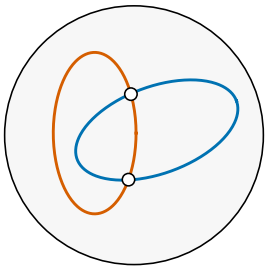


Figure 7: Two loops on a ball cross an even number of times.

Where do two zones cross? Exactly in the faces that belong to both, once in each. And every face is the crossing of exactly one pair of different zones. So F is the total number of crossings, added up over all pairs of zones, and that is a sum of even numbers. So F is even. This contradicts our assumption that F is odd, and the parallelograms are done. Rectangles, rhombi and squares are parallelograms, so they are done too.

> **Where the ball was used.** In the sentence "such a loop cuts the ball into two pieces". On a doughnut a loop can run round the ring without cutting it in two. On the doughnut of nine squares the zones are such loops: any two of them from different directions cross just once.

5. Kites: the plan

For kites the zones are useless: opposite sides of a kite are not parallel, a strip of kites bends, and nothing is carried along it. The argument that works is not a counting argument at all. It attaches a number to each face, in such a way that

- the number is negative for every face, and
- the product of the numbers of all the faces is positive, because of the ball.

A product of F negative numbers is positive only if F is even. This can be done at once if the polyhedron is flat, as we shall see in Section 6. For a polyhedron in space the "numbers" have to be matrices (Section 7), the reason for the positive product becomes a statement about nets on a ball (Section 8), and the faces first have to be reshaped so that their corners lie on circles (Section 9).

5.1. Two colours

One tool is needed throughout.

> **Two colours.** In a net on a ball in which every face has four sides, the vertices can be coloured black and white so that every edge joins a black vertex to a white one.

For the cube this is the familiar colouring of its eight corners (Figure 8). In general the reason is that every closed path along the edges of such a net has an even number of steps. We prove that in Section 8, as the first use of the main lemma there.

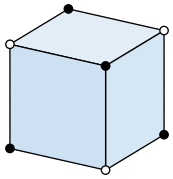


Figure 8: The two colours on a cube.

Going round a face, the colours alternate. So in a kite the corners A and T have one colour, and S and U have the other.

6. Flatland: the proof in a plane

Imagine a polyhedron squashed completely flat, so that all its vertices lie in one plane and its faces lie on top of each other like the two sides of a pillowcase. Suppose every face is a four-sided polygon whose corners lie on a circle. We show that the number of faces is even. This is a rehearsal: it contains the whole idea, and the arithmetic is that of complex numbers.

Treat the plane as the complex numbers, so that every vertex is a complex number z. Recall what division does: for three points z_1, z_2, z_3, the quotient

$$\frac{z_1 - z_2}{z_3 - z_2}$$

is a complex number whose angle is the angle at the corner z_2, measured from the direction of z_3 to the direction of z_1.

The sign of a face in the plane. Let z_1, z_2, z_3, z_4 be the corners, in order, of a quadrilateral whose sides do not cross, lying on a circle. Then the number

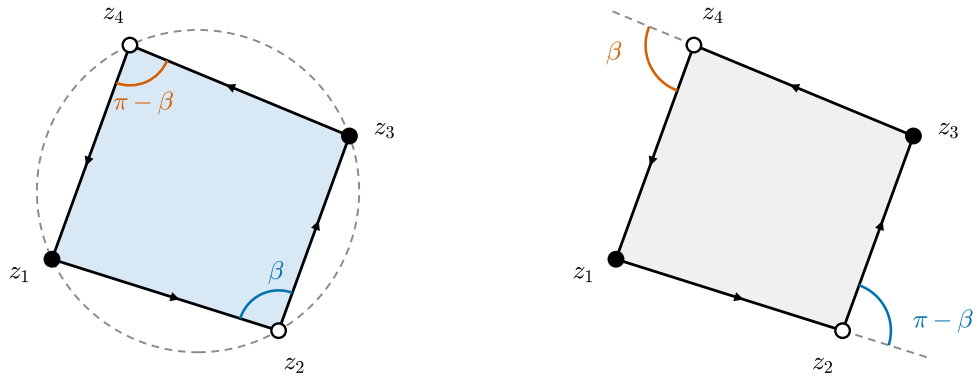
$$c = \frac{z_1 - z_2}{z_3 - z_2} \cdot \frac{z_3 - z_4}{z_1 - z_4}$$

is real and negative.

Why. Suppose first that the corners run counterclockwise. The first factor has the angle of the quadrilateral at z_2, and the second has its angle at z_4, both measured counterclockwise. When numbers are multiplied their angles add. A quadrilateral with its corners on a circle has the property that opposite angles add up to 180° (Figure 9, left); this is a school theorem that we take on trust. So c has angle 180°: it lies on the negative real axis. If the corners run clockwise, both angles are measured clockwise, they add up to −180°, and that is the same half-line. □

Now colour the vertices black and white, and give every edge a number: if its black end is x and its white end is y, the number is $y - x$. For a walk along the edges we make a rule.

Walking an edge from black to white, multiply by its number.
Walking it from white to black, divide by its number.



opposite angles add up to π — the turns at z_2 and z_4 add up to π

Figure 9: A quadrilateral with its corners on a circle. Left: the angles at z_2 and z_4. Right: someone walking round the boundary turns through $\pi - \beta$ at z_2 and through β at z_4, half a turn in all.

Walk round a face, counterclockwise as seen from outside the ball, starting at a black corner, and call the corners z_1, z_2, z_3, z_4 in the order of the walk. Then z_1 and z_3 are black, z_2 and z_4 white, and the rule says: multiply by $z_2 - z_1$, divide by $z_2 - z_3$, multiply by $z_4 - z_3$, divide by $z_4 - z_1$. The result is

$$\frac{z_2 - z_1}{z_2 - z_3} \cdot \frac{z_4 - z_3}{z_4 - z_1}.$$

Change the signs above and below in both fractions, and this is exactly the number c of the box. So the walk round every face produces a negative number. A face on the back of the flattened polyhedron is walked clockwise in the plane, and the box covers that too.

Multiply the numbers c of all the faces. Every edge is walked twice, in opposite directions (Section 2). So its number is multiplied in once and divided out once. Everything cancels, and the product of all the numbers c is 1.

But each c is negative. A product of F negative numbers that equals $+1$: F is even.

> **What to remember.** Each edge carries a number, white end minus black end. Walking the edge one way multiplies by it, walking it the other way divides by it. Round a face with corners on a circle the result is negative. Over the whole ball everything cancels, because every edge is walked once each way. In space the same sentences will be true with matrices in place of complex numbers. But complex numbers can be multiplied in any order and matrices cannot, so the cancellation will need a real argument. That argument is the ball lemma of Section 8.

7. Multiplying vectors

In the plane we could multiply and divide the edges, because points of the plane are complex numbers. Vectors in space cannot be multiplied in that way. But there is a way to turn vectors into things that can be multiplied, namely square matrices, and it works in every dimension d.

Mirror matrices. For every d there are square matrices $G_1, ..., G_d$, all of one size, such that

$$G_i G_i = I \quad \text{and} \quad G_i G_j = -G_j G_i \ \text{ for } \ i \neq j.$$

Here I is the identity matrix. For $d = 2$ one can take

$$G_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad G_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

and for $d = 3$ one more, with complex entries:

$$G_3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}.$$

It takes a minute to check the rules for these three by hand, and it is worth doing. For larger d the matrices are built from these by a doubling trick, which is given in Appendix A. Their size grows with d, but we shall never need to look inside them. Only the rules matter.

To a vector $v = (v_1, ..., v_d)$ we attach the matrix

$$\mathrm{Mat}(v) = v_1 G_1 + ... + v_d G_d.$$

Multiply two of these and use the rules. The terms with $i = j$ give $u_i v_i I$. The terms with $i \neq j$ come in pairs, $u_i v_j G_i G_j$ and $u_j v_i G_j G_i = -u_j v_i G_i G_j$. So

$$\mathrm{Mat}(u)\,\mathrm{Mat}(v) = (u \cdot v) I + \sum_{i<j} (u_i v_j - u_j v_i) G_i G_j.$$

The first term is the dot product. The second changes sign when u and v are exchanged. Hence

$$\mathrm{Mat}(u)\,\mathrm{Mat}(v) + \mathrm{Mat}(v)\,\mathrm{Mat}(u) = 2(u \cdot v) I. \tag{2}$$

Everything we need follows from (2).

- **Squares are lengths.** $\mathrm{Mat}(v)\,\mathrm{Mat}(v) = |v|^2 I$.
- **Perpendicular vectors anticommute.** If $u \cdot v = 0$ then $\mathrm{Mat}(u)\,\mathrm{Mat}(v) = -\mathrm{Mat}(v)\,\mathrm{Mat}(u)$.
- **Vectors can be divided by.** If $|v|^2 \neq 0$, the matrix $\mathrm{Mat}(v)$ has the inverse $\mathrm{Mat}(v)/|v|^2$.

And Mat is linear: $\mathrm{Mat}(su + tv) = s\,\mathrm{Mat}(u) + t\,\mathrm{Mat}(v)$ for numbers s and t.

> **The complex numbers are still there.** In the plane, put $J = G_1 G_2$. Then $JJ = G_1 G_2 G_1 G_2 = -G_1 G_1 G_2 G_2 = -I$: the matrix J is a square root of -1. For two vectors of the plane $u \cdot v = |u||v| \cos\theta$ and $u_1 v_2 - u_2 v_1 = |u||v| \sin\theta$, where θ is the angle from u to v. So the formula above says $\mathrm{Mat}(u)\,\mathrm{Mat}(v) = |u||v|(\cos\theta \cdot I + \sin\theta \cdot J)$: the product of two vectors of the plane is a complex number in disguise. The flat proof of Section 6 is the plane version of what follows.

7.1. A dot product with a weight

We shall need one variation. Fix a number τ, which may be negative, and for vectors with one extra coordinate, (x, s) and (y, t) with x, y in $\mathbb{R}^d$ and s, t numbers, define the *weighted dot product*

$$\langle (x, s), (y, t) \rangle = x \cdot y + \tau s t.$$

If $\tau = 1$ this is the ordinary dot product in $\mathbb{R}^{d+1}$. If τ is positive it is the ordinary dot product after the last coordinate has been rescaled. If τ is negative it is something new: a vector can then have a negative "squared length" $\langle v, v\rangle$.

The matrices adapt. Let ω be a square root of τ, a complex number if τ is negative, and put

$$\mathrm{Mat}(x, s) = x_1 G_1 + ... + x_d G_d + \omega s G_{d+1}.$$

The same computation as before gives rule (2) with the weighted dot product in place of the ordinary one, because $\omega^2 = \tau$. With it come the three consequences: $\mathrm{Mat}(v)\,\mathrm{Mat}(v) = \langle v, v\rangle I$, vectors with $\langle u, v\rangle = 0$ anticommute, and $\mathrm{Mat}(v)$ can be divided by when $\langle v, v\rangle \neq 0$. Nothing else about τ will be used, so its sign will not matter.

8. Signs on a ball

For kites, this section contains the one place where the shape of the surface is used, apart from Euler's formula in Section 3. Forget polyhedra for a moment and consider only a net on a ball.

Suppose every edge carries an invertible matrix, in the following sense: for each edge, and each of the two directions in which it can be walked, a matrix is given, and the matrices for the two directions are inverse to each other. For a path along the edges, multiply the matrices of its steps, writing each new one to the left of the earlier ones. Call the result the *product along the path.*

A matrix of the form cI, a number times the identity, is called a *scalar matrix.* Scalar matrices commute with all matrices.

The ball lemma. Suppose that the product round every face f, walked counterclockwise, is a scalar matrix $c_f I$. Then

(i) the product along every closed path is a scalar matrix, and its scalar is a product of numbers c_f and their inverses;

(ii) the product of the numbers c_f over all faces is 1.

Before the proof, three remarks. First, when the product round a face is scalar, it does not matter at which corner the walk starts. Starting one step later changes the product B into HBH^{-1}, where H is the matrix of the first step, and $H(cI)H^{-1} = cI$. Second, walking the face the other way round gives the inverse, $c^{-1}I$. Third, statement (ii) is exactly the cancellation we saw in Flatland, but there it was free, because numbers commute.

Why. We use induction on the number of faces.

One face. Call a closed path a *loop* if it visits no vertex twice. A net with one face has no loop: a loop would cut the ball into two pieces, and each piece would contain a face. Now take any closed path in such a net. If it never walked along an edge and straight back, then the part of it between the two visits to the first vertex that it visits twice would be a loop. So somewhere it walks along an edge and comes straight back. Those two steps multiply to I. Delete them and repeat, until nothing is left. So the product along every closed path is I. That is (i). The boundary of the one face is a closed path, so $c_f = 1$. That is (ii).

More than one face. Somewhere there is an edge e with two different faces f and f' on its two sides: walk on the ball from the inside of one face to the inside of another, and look at the edge

you cross at the moment you first enter a different face. Say e runs from x to y, and write H for its matrix in that direction. The walk round f can be taken as: first e, then a path L from y back to x. The walk round f' uses e in the opposite direction: first e backwards, then a path R from x back to y (Figure 10). Neither L nor R uses e, because e has f on one side only and f' on the other side only. Writing $[L]$ and $[R]$ for the products along L and R,

$$c_f I = [L]H, \quad c_{f'} I = [R]H^{-1}.$$

Now erase the edge e from the net. The net stays connected, because L still joins x and y. The two faces merge into one, which is again a disc (the third fact about the ball that we take on trust), and the walk round it is L followed by R. Its product is

$$[R][L] = \big(c_{f'} H\big)\big(c_f H^{-1}\big) = c_f c_{f'} I.$$

So the new net again satisfies the assumption of the lemma, it has one face fewer, and the product of its face numbers is the same as before. By induction, (ii) holds for the new net, so it holds for the old one.

For (i), take a closed path in the old net. Wherever it walks along e, let it walk along L backwards instead, which also leads from x to y. Since $H = [L]^{-1} c_f$, this changes the product only by the factor $1/c_f$. Do the same, with L forwards, wherever the path walks along e backwards. The new path avoids e, so it is a closed path in the new net, and (i) holds for it by induction. □

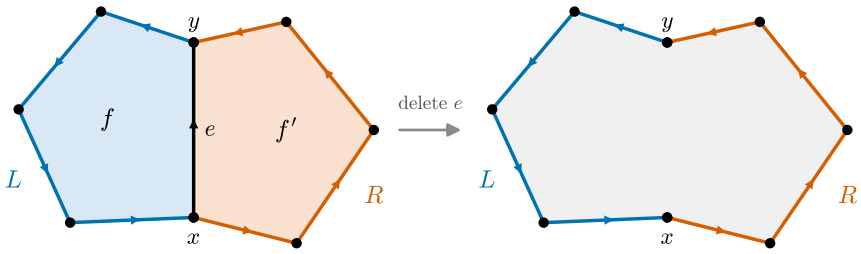


Figure 10: Erasing an edge between two faces merges them into one.

8.1. Two uses straight away

Two colours. Take a net in which every face has four sides, and put the 1×1 matrix (-1) on every edge, in both directions. Round every face the product is $(-1)^4 = 1$. By (i), the product along every closed path is 1. So every closed path has an even number of steps. Now colour one vertex black, and colour every other vertex black or white according to whether an even or an odd number of steps leads to it. Different paths to the same vertex give the same answer, since together they form a closed path. This proves the two-colour statement of Section 5.1.

Heights. Suppose every edge carries a number κ for one direction and $-\kappa$ for the other, and the numbers add up to 0 round every face. Put the matrix $\left(\begin{smallmatrix} 1 & \kappa \\ 0 & 1 \end{smallmatrix}\right)$ on the edge. These matrices multiply by adding their corners:

$$\begin{pmatrix} 1 & \kappa \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & \kappa + \lambda \\ 0 & 1 \end{pmatrix}.$$

So the product round every face is I, and by (i) the product along every closed path is I: **the numbers add up to 0 along every closed path.** Then we can define a *height* for every vertex.

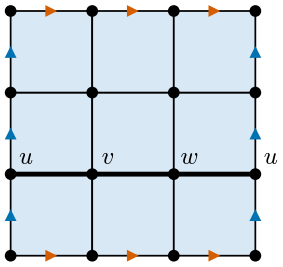


Figure 11: The net of the nine-square doughnut. The left side is glued to the right side and the bottom to the top. The thick path closes up after three steps.

Give one vertex the height 0, and give every other vertex the sum of the numbers along a path leading to it. Again the path does not matter. The number on an edge is then the difference of the heights of its two ends.

8.2. Why the doughnut escapes

On the doughnut made of nine squares, draw the net as a 3×3 board whose opposite sides are glued (Figure 11). The path $u \to v \to w \to u$ along one row is closed and has three steps. So there is no colouring with two colours, although every face has four sides, and the ball lemma must fail on a doughnut. Where does its proof break? Erasing edges works on a doughnut too. It is the last step that fails: on a doughnut a net with one face still has loops, like the thick path in the figure, because a loop need not cut a doughnut in two.

9. The stretch

We return to a polyhedron whose faces are congruent kites or darts, with shape number r different from 0 and 2, and with labels A, S, T, U as in Figure 5. Its edges are of two kinds: the a-edges, of length a, and the b-edges, of length b. The vertices are coloured black and white.

9.1. Heights

Give every edge a number by the following rule (Figure 12):

walking from its black end to its white end, the height changes by
$+r$ along an a-edge, and by $-(2-r)$ along a b-edge.

Walking from white to black, the changes are the opposite ones. The two amounts and their signs are chosen so that the two things we are about to check come out right: the changes cancel round every face, and the heights obey the kite rule. Check a face in which A is black, walking $A \to S \to T \to U \to A$. The four changes are

$$+r, \quad +(2-r), \quad -(2-r), \quad -r,$$

and they add up to 0. If A is white, all four signs flip. By the second use of the ball lemma there is a height η_v for every vertex v with these differences. In a face with A black,

$$\eta_S = \eta_U = \eta_A + r, \quad \eta_T = \eta_A + 2,$$

and with A white the same with $-r$ and -2.

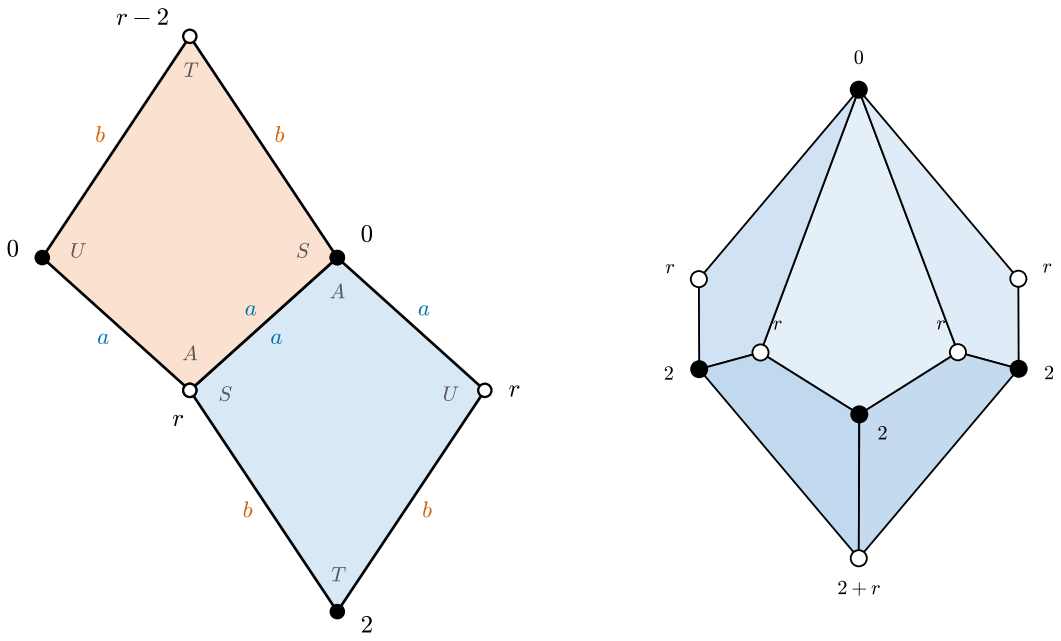


Figure 12: Heights. Left: two neighbouring faces, one with A black and one with A white. Right: on the solid with ten kites the heights are 0, r, 2 and $2+r$ from one tip to the other. Its a-edges are the long edges at the two tips, and its r is the golden ratio, 1.618...

The heights obey the kite rule. In both cases $\eta_S + \eta_U = (2-r)\eta_A + r\eta_T$. For A black the left side is $2\eta_A + 2r$ and the right side is $(2-r)\eta_A + r(\eta_A + 2)$, which is the same.

9.2. One more coordinate

Use the height as a new coordinate. For every vertex put

$$\hat{X}_v = (X_v, \eta_v),$$

a point with one coordinate more than before. Since the positions X_v and the heights η_v both obey the kite rule, so do the new points:

$$\hat{X}_S + \hat{X}_U = (2-r)\hat{X}_A + r\hat{X}_T.$$

As in Section 3 this says that the midpoint of $\hat{X}_S$ and $\hat{X}_U$ lies on the line through $\hat{X}_A$ and $\hat{X}_T$. So the four new corners of a face still lie in one plane. **The lifted polyhedron is again built from flat four-sided faces.** But the faces have a new shape, and how much the shape changes depends on how strongly the new coordinate counts. That is what the weight τ of Section 7.1 is for: we measure the lifted polyhedron with the weighted dot product $\langle (x,s),(y,t)\rangle = x \cdot y + \tau st$, and we are free to choose τ.

9.3. Choosing the weight

We want a right angle at the corner S. The two edges at S, as vectors pointing away from S, are $\hat{X}_A - \hat{X}_S$ and $\hat{X}_T - \hat{X}_S$. Their new coordinates are $\eta_A - \eta_S = \mp r$ and $\eta_T - \eta_S = \pm(2-r)$, with the upper signs if A is black. So

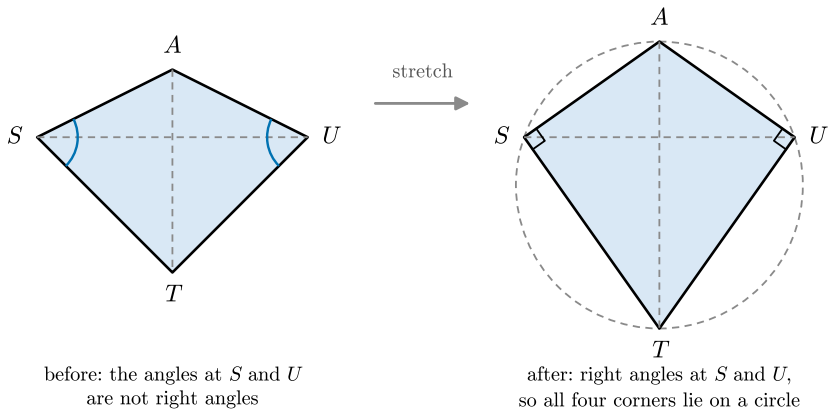


Figure 13: One face before and after the lift, when τ is positive. This is the kite of the worked example.

$$\left\langle \hat{X}_A - \hat{X}_S, \hat{X}_T - \hat{X}_S \right\rangle = (X_A - X_S) \cdot (X_T - X_S) - \tau r(2-r).$$

This is zero if we choose

$$\tau = \frac{(X_A - X_S) \cdot (X_T - X_S)}{r(2-r)}.$$

The numerator is a dot product of two edge vectors of one face. All faces are congruent, so it is the same for every face: **one choice of τ gives a right angle at S in every face at once.** A kite is symmetric, so the same τ gives a right angle at U.

What has happened to a face? Its diagonal SU has not changed, because S and U have the same height. Its other diagonal AT, along the axis, has a new length, because A and T differ in height by 2: its squared length was $|AT|^2$ and is now $|AT|^2 + 4\tau$. So if τ is positive the face has been stretched along its axis, by just the amount that makes the angles at S and U right angles (Figure 13). And a quadrilateral with right angles at two opposite corners has its four corners on a circle, namely the circle with diameter AT; this is the second school theorem that we take on trust.

> **A worked example.** Take the kite with corners $S = (-1, 0)$, $U = (1, 0)$, $A = \left(0, \frac{1}{2}\right)$ and $T = (0, -1)$. Here $M = (0, 0)$ is one third of the way from A to T, so $r/2 = 1/3$ and $r = 2/3$. The edge vectors at S are $A - S = \left(1, \frac{1}{2}\right)$ and $T - S = (1, -1)$, with dot product $\frac{1}{2}$, and $r(2-r) = \frac{8}{9}$. So $\tau = \frac{9}{16}$. In a face with A black the heights are 0 at A, $\frac{2}{3}$ at S and U, and 2 at T. The lifted edges at S are $\left(1, \frac{1}{2}, -\frac{2}{3}\right)$ and $\left(1, -1, \frac{4}{3}\right)$, and their weighted dot product is $1 - \frac{1}{2} - \frac{9}{16} \cdot \frac{2}{3} \cdot \frac{4}{3} = 0$. The axis had length $\frac{3}{2}$, and its new squared length is $\frac{9}{4} + 4 \cdot \frac{9}{16} = \frac{9}{2}$.

9.4. The new edge lengths

Two numbers will be needed: the weighted squared lengths of the lifted edges. Write σ for the length of the diagonal SU. A computation with the Pythagorean theorem, given in Appendix B, shows that

$$N_a = \frac{\sigma^2}{2(2-r)} \quad \text{for every } a\text{-edge}, \qquad N_b = \frac{\sigma^2}{2r} \quad \text{for every } b\text{-edge}.$$

All that matters is that these are two fixed numbers, the same in every face, and that neither is zero. In the worked example $\sigma = 2$, so $N_a = \frac{3}{2}$ and $N_b = 3$.

9.5. When the weight is negative

The formula for τ can give a negative number. That happens when the angle at S is larger than a right angle, so that the axis would have to be shortened, and it always happens for darts. The solid with ten kites in Figure 12 is an example: as drawn, its angles at S are about 102°. Then the "stretch" cannot be drawn. The weighted dot product is not a dot product of ordinary space, there are vectors with negative squared length, and for darts N_a or N_b is negative.

This does no harm. From here on the proof never looks at a picture. It uses the three rules for the matrices $\mathrm{Mat}(v)$ from Section 7.1, which hold for every τ, and the three facts we have just established:

(a) the lifted corners of every face obey the kite rule;
(b) $\left\langle \hat{X}_A - \hat{X}_S, \hat{X}_T - \hat{X}_S \right\rangle = 0$ in every face;
(c) every lifted a-edge has weighted squared length N_a, and every lifted b-edge has N_b, and these are not zero.

When τ is positive you may picture a real polyhedron with faces on circles. When it is negative, the same formulas are true and the same conclusion follows.

10. The sign of a kite, and the end of the proof

Fix a face and walk round it: $A \to S \to T \to U \to A$. The four lifted edge vectors along the walk are

$$p = \hat{X}_S - \hat{X}_A, \quad q = \hat{X}_T - \hat{X}_S, \quad p' = \hat{X}_U - \hat{X}_T, \quad q' = \hat{X}_A - \hat{X}_U.$$

Write P, Q, P', Q' for their matrices $\mathrm{Mat}(p), \mathrm{Mat}(q), \mathrm{Mat}(p'), \mathrm{Mat}(q')$.

The sign of a kite. $Q'P'QP = -N_aN_bI$.

Why. From fact (c), $PP = N_aI$ and $QQ = N_bI$. From fact (b), p and q have weighted dot product zero, so $QP = -PQ$.

From fact (a) we can express p' and q' through p and q. The kite rule gives $\hat{X}_U = (2-r)\hat{X}_A + r\hat{X}_T - \hat{X}_S$. Substituting this,

$$p' = -(2-r)p - (1-r)q, \quad q' = (1-r)p - rq,$$

as you can check by writing out both sides. Since Mat is linear, the same holds for the matrices. Now multiply out, using $PP = N_aI$, $QQ = N_bI$ and $QP = -PQ$:

$$\begin{aligned} Q'P' &= ((1-r)P - rQ)(-(2-r)P - (1-r)Q) \\ &= [-(1-r)(2-r)N_a + r(1-r)N_b]I - [(1-r)^2 + r(2-r)]PQ. \end{aligned}$$

Put in the values of N_a and N_b. The first bracket is $-(1-r)\sigma^2/2 + (1-r)\sigma^2/2 = 0$. The second bracket is $1 - 2r + r^2 + 2r - r^2 = 1$. So $Q'P' = -PQ$, and therefore $Q'P'QP = -PQQP = -N_bPP = -N_aN_bI$. □

10.1. Matrices on the edges

Now put a matrix on every edge, as in Flatland: white end minus black end. For an edge with black end x and white end y let

$$H = \mathrm{Mat}\big(\hat{X}_y - \hat{X}_x\big).$$

By fact (c), HH is $N_a I$ or $N_b I$, which is not zero, so H is invertible. The rule is the one of Flatland: walking the edge from black to white, multiply by H. Walking it from white to black, multiply by H^{-1}.

Walk round a face in which A is black, in the order $A \to S \to T \to U \to A$.

- $A \to S$ is black to white. The matrix is P.
- $S \to T$ is white to black. The edge's matrix is $\mathrm{Mat}\big(\hat{X}_S - \hat{X}_T\big) = -Q$, and we multiply by its inverse, $-Q/N_b$.
- $T \to U$ is black to white. The matrix is P'.
- $U \to A$ is white to black. We multiply by the inverse of $-Q'$, which is $-Q'/N_a$.

The product along the walk, later steps on the left, is

$$\left(-\frac{Q'}{N_a}\right)P'\left(-\frac{Q}{N_b}\right)P = \frac{Q'P'QP}{N_a N_b} = -I.$$

If A is white, the four steps contribute $-P/N_a$, Q, $-P'/N_b$ and Q', and the product is the same. If the walk $A \to S \to T \to U$ happens to be clockwise, the counterclockwise walk gives the inverse of $-I$, which is $-I$ again.

So the product round every face is $-I$. This is a scalar matrix, with $c_f = -1$. The ball lemma says that the product of the numbers c_f over all faces is 1:

$$(-1)^F = 1.$$

So F is even. This contradicts the assumption that F is odd, and the theorem is proved for kites and darts.

> **What is this sign?** This aside is for orientation only; the proof does not use it. The matrix $\mathrm{Mat}(v)$ of a vector stands for the mirror reflection that reverses v, and a product of two of them stands for a rotation through twice the angle between the two vectors. Someone walking round a face with its corners on a circle turns through half a turn in all at the two white corners (Figure 9, right). So the matrices record a rotation through one full turn. A full turn brings every vector back to where it was. But these matrices remember a full turn as $-I$, and only two full turns as I. Physicists know this behaviour from the electron, and it can be demonstrated with a twisted belt.

11. The last case: a corner of 180 degrees

One case is left: $r = 0$ or $r = 2$. Take $r = 0$. Then A is the midpoint of SU. The face is a triangle STU, and its base is made of two edges of the net, SA and AU. These half-bases are the edges of length a, so each half-base is glued to a half-base of another triangle.

Colour the vertices black and white. For each half-base, take the vector that points from its black end to its white end. In one face the two halves AS and AU point in opposite directions away from A, and A is the black end of both or the white end of both. So their two vectors are opposite: one is the negative of the other.

Every face has two half-bases, and every half-base belongs to two faces. So, as with the zones in Section 4, we can step from a face across a half-base into the next face, leave it across its other half-base, and so on. The walk can be traced backwards in only one way, so it closes up where it

started. The faces fall into closed chains (Figure 14 shows part of one). Along a chain the vector flips at every face. After a full round of k faces we are back at the half-base we started with, and its vector has been multiplied by $(-1)^k$. A nonzero vector is not its own negative. So k is even. Every chain has an even number of faces, and so F is even.

For $r = 2$ the same argument works with T in place of A and the two b-edges as the half-bases.

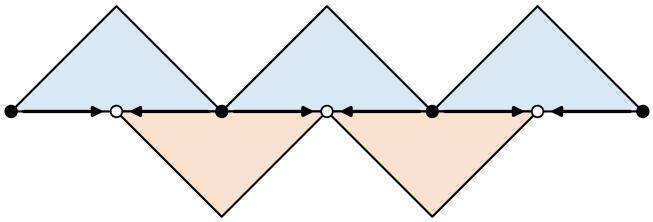


Figure 14: Part of a chain of triangles glued along halves of their bases, seen flat. The arrows on the half-bases point from black to white, and they alternate.

This completes the proof of the theorem.

12. Looking back

What was used. From the faces we used the kite rule and the fact that they are congruent. The paper for mathematicians shows that less is enough: the kite rule and one common length, the diagonal SU. From the ball we used Euler's formula, the fact that a loop cuts it in two, and the ball lemma, three times: for the two colours, for the heights, and for the final product of signs.

A more general fact. The same method proves more, though we do not carry it out here: *if a closed surface without holes is built from four-sided faces, each with its corners on a circle and without crossed sides, then the number of faces is even.* The faces need not be congruent. For such a face the product round it is a negative number times I, and by the ball lemma these F negative numbers multiply to 1. Faces with a larger even number of corners on a circle can be treated in the same way. Those with $4, 8, 12, ...$ sides give a negative number and those with $6, 10, 14, ...$ sides a positive one, and the conclusion is that the surface has an even number of vertices.

What was known. The step from Flatland to the ball lemma has close relatives in recent mathematics. Sergey Fomin and Pavlo Pylyavskyy (2023) and Anton Izosimov (2026) multiply such numbers round the faces of a surface to prove theorems about points on circles: on a surface built from four-sided faces, if the corners of all faces but one lie on circles, so do the corners of the last. In these circle theorems the sign plays no part, and no faces are counted. What is new here is the count that the sign gives, and above all the stretch, which brings identical kites within its reach.

The doughnut. In six-dimensional space there is a doughnut-shaped surface built from 27 identical squares whose vertices can even be coloured with two colours. Read each square as a kite with $r = 1$, with A and T at opposite corners. Then every step of Section 9 and Section 10 works on it, the product round every face is $-I$, and 27 is odd. Only the ball lemma fails. We do not know whether a doughnut with an odd number of identical faces exists in ordinary three-dimensional space.

How the proof was found. The problem was studied for four days in October 2026 by AI systems directed by the same person. That work reduced it to kites, found the height function and the stretch, and tested many approaches that failed. It was then set aside. On 12 November 2026 a newer system found the missing step, the sign of a face. A paper written for mathematicians accompanies this guide.

Appendix A: the mirror matrices for every dimension

Suppose we have d matrices $G_1, ..., G_d$ of size $N \times N$ that obey the rules $G_k G_k = I$ and $G_k G_l = -G_l G_k$ for $k \neq l$. Then the following $d+2$ matrices of size $2N \times 2N$, written in blocks, obey them too:

$$\begin{pmatrix} 0 & G_k \\ G_k & 0 \end{pmatrix} \text{ for } k = 1, ..., d, \quad \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \quad \begin{pmatrix} 0 & -iI \\ iI & 0 \end{pmatrix}.$$

Here i is the imaginary unit. Blocks multiply like the entries of 2×2 matrices, as long as the order of the factors is kept. For example

$$\begin{pmatrix} 0 & G_k \\ G_k & 0 \end{pmatrix} \begin{pmatrix} 0 & G_l \\ G_l & 0 \end{pmatrix} = \begin{pmatrix} G_k G_l & 0 \\ 0 & G_k G_l \end{pmatrix},$$

which is I for $k = l$ and changes sign when k and l are exchanged. The other checks are of the same kind. Start with the single 1×1 matrix (1). One doubling gives the three 2×2 matrices of Section 7, the next gives five matrices of size 4×4, and so on. For an even d, leave one matrix out.

Appendix B: the lengths of the lifted edges

Let α be the length of the axis AT of the original kite and σ the length of SU. The two diagonals are perpendicular and meet at the midpoint M of SU. Since $M = A + (r/2)(T - A)$, the vectors from M to A and from M to T are $-(r/2)(T - A)$ and $(1 - r/2)(T - A)$. The vectors from M to S and to U have length $\sigma/2$ and are perpendicular to these. By the Pythagorean theorem

$$a^2 = \frac{\sigma^2}{4} + \frac{r^2\alpha^2}{4}, \quad b^2 = \frac{\sigma^2}{4} + \frac{(2-r)^2\alpha^2}{4}.$$

For the dot product of the vectors from S to A and from S to T, go from S to M first: each vector is the vector from S to M plus a vector along the axis. The mixed terms vanish, because the diagonals are perpendicular, and what remains is $\sigma^2/4 - (r/2)(1 - r/2)\alpha^2$. So

$$\tau = \frac{\sigma^2 - r(2-r)\alpha^2}{4r(2-r)}.$$

Along an a-edge the height changes by $\pm r$, so its weighted squared length is

$$N_a = a^2 + \tau r^2 = \frac{\sigma^2}{4} + \frac{r^2\alpha^2}{4} + \frac{r\sigma^2}{4(2-r)} - \frac{r^2\alpha^2}{4} = \frac{\sigma^2}{2(2-r)}.$$

Along a b-edge the height changes by $\pm(2 - r)$, and in the same way $N_b = b^2 + \tau(2-r)^2 = \sigma^2/(2r)$. The lifted axis has weighted squared length $\alpha^2 + 4\tau = \sigma^2/(r(2-r))$: this is the square of the stretched length of the axis in Figure 13.